EXPERIMENT NO: 2

STUDY OF LAN CONFIGURATION

AIM

To configure a Local Area Network (LAN) by connecting computers through a switch and verify connectivity between them.

REQUIREMENTS

5 PCs, 1 switch, Ethernet cables.

IP ADDRESSING

PC	IP Address	Subnet Mask
PC1	10.20.30.1	255.255.255.0
PC2	10.20.30.2	255.255.255.0
PC3	10.20.30.3	255.255.255.0
PC4	10.20.30.4	255.255.255.0
PC5	10.20.30.5	255.255.255.0

PROCEDURE

1. Connect all PCs to the switch using Ethernet cables.
2. Open Network Settings on each PC and set the IP address and subnet mask as per the table.
3. Open Command Prompt and check the configuration using ipconfig.
4. Ping other PCs to check connectivity.

OUTPUT

```
C:\>ping 10.20.30.2

Pinging 10.20.30.2 with 32 bytes of data:
Reply from 10.20.30.2: bytes=32 time<1ms TTL=128
Reply from 10.20.30.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.20.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

RESULT

The LAN was configured successfully and connectivity between the PCs was verified using ping.